

BIOS vs UEFI Comparison

ITEP 414 – System Administration and Maintenance | Week 3 Portfolio Project

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Comparison Table

Aspect	BIOS	UEFI
Definition	Legacy firmware standard from the 1980s that initializes hardware before OS handoff.	Modern firmware standard designed to replace BIOS with more features and flexibility.
Boot Process	Reads the first sector (MBR) and executes a 16-bit boot loader.	Reads .efi files directly from an EFI System Partition; supports 32/64-bit.
Maximum Disk Support	2 TB (MBR partitioning limit)	Effectively unlimited (GPT partitioning, 9.4 ZB theoretical)
Partition Style	MBR, max 4 primary partitions	GPT, up to 128 partitions
Security Features	None built in	Secure Boot, cryptographic verification of boot loaders
Speed	Slower boot (16-bit real mode, sequential device init)	Faster boot (32/64-bit mode, parallel init)
Advantages	Simple, broad legacy compatibility	Faster boot, larger disks, Secure Boot, GUI, network support
Disadvantages	Slow, 2TB limit, no security features, 16-bit only	More complex firmware; legacy OSes may need compatibility mode
Modern Usage	Legacy systems, some older enterprise hardware	Standard on all new PCs and servers since ~2012

Why UEFI Has Largely Replaced BIOS

UEFI has become the standard firmware in modern computing because it directly addresses BIOS's core limitations. Its use of GPT partitioning removes the 2 TB disk-size and four-partition ceilings that constrain BIOS/MBR systems, which matters for enterprise servers with large storage arrays. UEFI also boots faster because it can initialize hardware in parallel rather than the sequential process BIOS relies on, and it operates in 32-bit or 64-bit mode instead of being limited to 16-bit real mode.

Security is another major driver of the shift. UEFI's Secure Boot feature cryptographically verifies boot loaders and OS kernels before execution, helping prevent rootkits and other boot-level malware from persisting on a system — a capability BIOS simply does not have. UEFI additionally supports richer pre-boot environments, including graphical interfaces, mouse input, and network booting, which simplify diagnostics and remote provisioning. For enterprise and cloud deployments in particular, UEFI's combination of larger disk support, faster boot, and built-in security make it the practical default, and most current operating systems — including Ubuntu Server — are built and tested primarily against UEFI firmware.